LOCAL COHOMOLOGY COMMUTES WITH BASE CHANGE ALONG A RING MAP FOR FINITELY GENERATED IDEALS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of local cohomology through the alternating Cech complex attached to generators of the ideal. Invoked near the end of the counted proof (source lines 1603-1606). Identifies the two complexes as representatives of the two sides.
- The statement on representing objects by K-flat complexes. Used inside the counted proof (source lines 1598-1601). Allows the passage to the level of complexes.

Source reference. Original lemma lemma-local-cohomology-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $A \to B$ be a ring homomorphism and let $I \subset A$ be a finitely generated ideal. Set $J = IB$. Let $Z = V(I)$ and $Y = V(J)$. Then*

$$R\Gamma_Z(K) \otimes^{\mathbf{L}}_A B = R\Gamma_Y(K \otimes^{\mathbf{L}}_A B)$$

functorially in $K \in D(A)$.

Proof. Write $I = (f_1, \ldots, f_r)$. Then J is generated by the images $g_1, \ldots, g_r \in B$ of $f_1, \ldots, f_r$. Then we have

$$(A \to \prod A_{f_{i_0}} \to \ldots \to A_{f_1 \ldots f_r}) \otimes_A B = (B \to \prod B_{g_{i_0}} \to \ldots \to B_{g_1 \ldots g_r})$$

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as complexes of B-modules. Represent K by a K-flat complex $K^\bullet$ of A-modules. Since the total complexes associated to

$$K^\bullet \otimes_A (A \to \prod A_{f_{i_0}} \to \ldots \to A_{f_1 \ldots f_r}) \otimes_A B$$

and

$$K^\bullet \otimes_A B \otimes_B (B \to \prod B_{g_{i_0}} \to \ldots \to B_{g_1 \ldots g_r})$$

represent the left and right hand side of the displayed formula of the lemma (see Lemma lemma-local-cohomology-adjoint) we conclude. $\square$